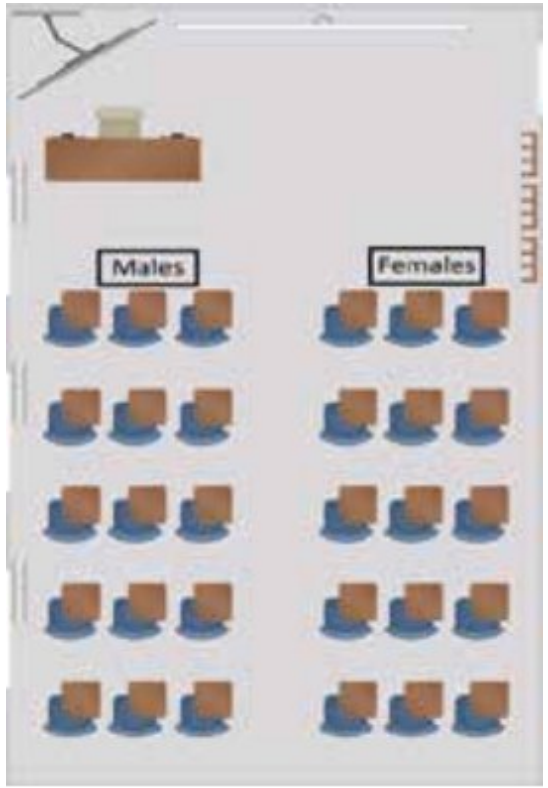
SWE 4101

Introduction to Software Engineering

Lec - 1

Classroom Rules



- Male and female students will sit in different columns/rows in the classroom. No male and female students can seat side-by-side.
- In case of individual sitting arrangement parted (tablet-chair), they may seat “next” to one another.
- If the number of female students is less than that of the male students, female students may seat in the front rows or front desks of the column. The rest of the seats located in the backside could be filled by the male students

✓ Be on time (If you're early, you're on time. If you're on time, you're late)

✓ Be a good Listener

✓ Ask questions

✓ Share new ideas

✓ Contribute to discussions

✓ No unethical Request

Google Classroom & Online Class Info

Join using Class Code

jssefg3q

Use only IUT email to join

Shohel Ahmed

Professional Experience:

Islamic University of Technology (IUT)
Assistant Professor (Feb'2020 ~)

Military Institute of Science & Technology
Assistant Professor (Sep'19 – Jan'20)

Kona Software Lab Ltd.
Sr. Project Manager (Jul'16 ~ Sep'19)

Samsung R&D Institute Bangladesh (SRBD)
Chief Engineer & Group Leader (Jan'15~ Jul'16)
Digital Imaging
Technical Lead (Dec'10~ Dec'14)
Platform Lab

Sylhet International University
Assistant Professor (Jul'09 – Nov'10)
Lecturer (Apr'04 – Jul'04)

Inha University, Korea
Research Assistant (Aug'07-Jun'09)
Intelligent Mobile Lab

Educational Experience:

Master of Engineering in Information Engineering
Inha University, Incheon, South Korea

BSc in Computer Science & Information Technology
Islamic University of Technology, Gazipur, Bangladesh

Higher Secondary Certificate (HSC)
Secondary School Certificate (SSC)
Rangpur Cadet College, Rangpur, Bangladesh

Patents:

- ✓ Shohel ahmed, Shake Md. Riazul Islam, Kwak Kyung Sup - " METHOD AND SYSTEM FOR FINGERPRINT-BASED PAYMENT " Application No.(Date) 10201701413 75 (2017.10.27),Publication : KR101951125B1

- ✓ BARI, A.S.M. Hossain, Shohel AHMED ,AHSAN, Shegufta Bakht, Sadre Ala PARVEZ,AZIZ, Syeda Persia, Md. Tawhidul Islam CHOWDHURY- "ELECTRONIC DEVICE AND METHOD FOR PROVIDING INFORMATION THEREOF "
  Publication number: 20150220247

For More info https://cse.iutoic-dhaka.edu/profile/sohel/

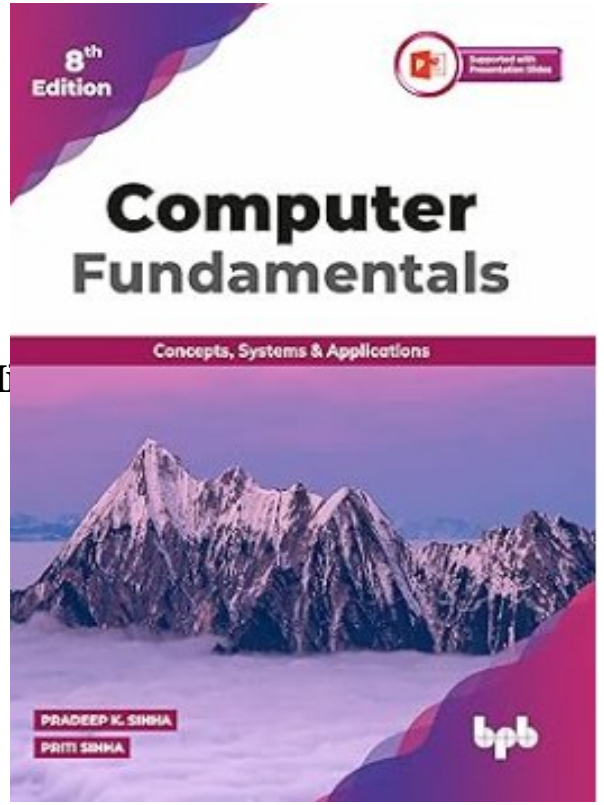
Important For you

Course Code: SWE 4101

Course Name: Introduction to Software Engineering

Recommended Books:

- P. K. Sinha and P. Sinha, *Computer Fundamentals*, 8th ed. BPB Publications, 2020
- R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*, 9th ed. McGraw-Hi



Following the lecture slides & class will be good enough

Assessment & Grading Policy

Course Assessment Method			
10% participation	30% Quizzes and assignments	30% Midterm	30% final

Grading Policy

Marks out of 100	Letter Grade	Grade Points	Marks out of 100	Letter Grade	Grade Point
80 - 100	A+	4.00	55 - 59	B-	2.75
75 - 79	A	3.75	50 - 54	C+	2.50
70 - 74	A-	3.50	45 - 49	C	2.25
65 - 69	B+	3.25	40 – 44	D	2.00
60 - 64	B	3.00	00 - 39	F	0.00

Course Objective

Upon completion of this course, you will be able to:

01 Understand basic concepts of computer systems, hardware, software, and data representation.

02 Explain computer organization, networking, and emerging computing technologies.

03 Demonstrate understanding of software engineering principles, processes, and maintenance.

Course Outcome(CO)

01 Understand the core components of a computer system - hardware, software, and their interaction.

02 Explain basic computer organization, networking, and web technologies with real-world applications.

03 Illustrate key stages of the software life cycle and core software engineering concepts.

Outcome Based Education (OBE)

Exam Result is Not the Most Important Consideration by Employer

- Communication (verbal & written)
- Honesty/Integrity
- Teamwork skills
- Interpersonal skills
- Strong work ethics
- Motivation & initiative
- Flexibility/adaptability
- Analytical skills
- Computer skills
- Creativity
- Organizational skills
- Detail oriented
- Leadership skills
- Self confidence
- Friendly/outgoing personality
- Well mannered / polite
- Tactfulness
- CGPA
- Sense of humor
- Entrepreneurial skills/risk taker

Lecture Plan

Weeks	Topics
1	Introduction to Computers and Basic Concepts
2	Hardware Components and Peripherals
3	Computer Architecture & Number Systems
4	Software Concepts – Types and System Software
5	Programming and Language Translators
6	Networking and the Internet
7	Web and Emerging Technologies
8	Introduction to SW Engineering, Software Development Life Cycle (SDLC)
9	Software Process Models
10	Software Requirements Engineering, UI/UX
11	Basics of SW design
12	SW Testing
13	Software Documentation
14	Software Release and Maintenance

Meet Your Digital Brain: The Computer



Desktop



Laptop



Mobile



ATM



Point of Sale (POS)



Samsung Gear 360

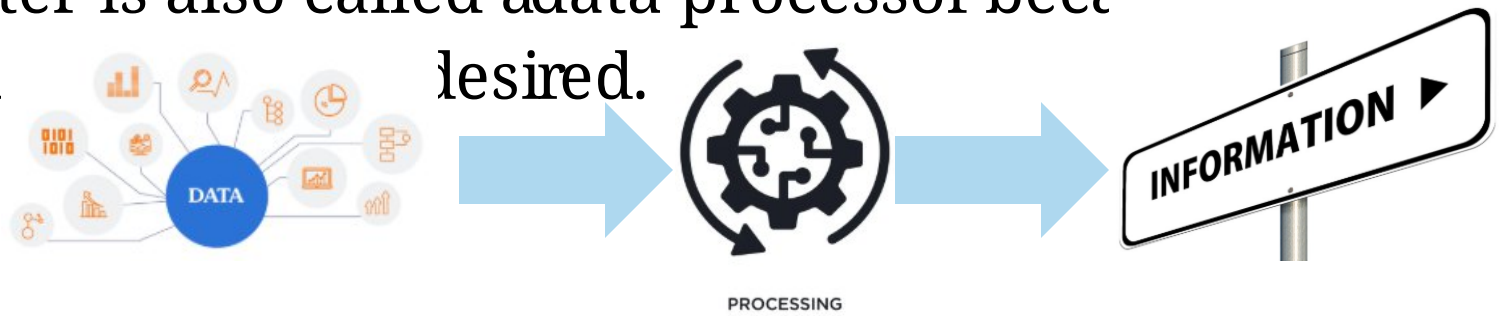


Samsung NX1

- ✓ A computer is an electronic device thatakes input, processes it, stores data, and produces outputto help us perform tasks.
- ✓ Your smartphone today is millions of times more powerful than computers from the 1960s!

Computer

- ✓ The word computer comes from the word “compute” which means, “to calculate”.
- ✓ Thereby, a computer is an electronic device that can perform arithmetic operations at high speed.
- ✓ A computer is also called a data processor because it can store, process, and retrieve d lesired.



The activity of processing data using a computer is called data processing

What are the activities?	Capture Data	Manipulate Data	Output Results

Data vs Information

Suppose IUT wants to understand the patterns of students arriving late to classes so that we can take steps to improve punctuality. You are asked to design a system to track late arrivals throughout the semester. What data should be collected about students arriving late? What information can be generated from that data to help the IUT authority?

Data can be collected:

- ✓ Student ID
- ✓ Date
- ✓ Time of arrival
- ✓ Course Code
- ✓ Teacher's name
- ✓ Reason for being late

Information that can be generated from the data:

- ✓ Students who are frequently late
- ✓ Average delay per student or per class
- ✓ Days or subjects with the highest late arrivals
- ✓ Patterns over weeks or months (e.g., mornings vs. afternoons)
- ✓ Most/Least Frequent cause of late arrival

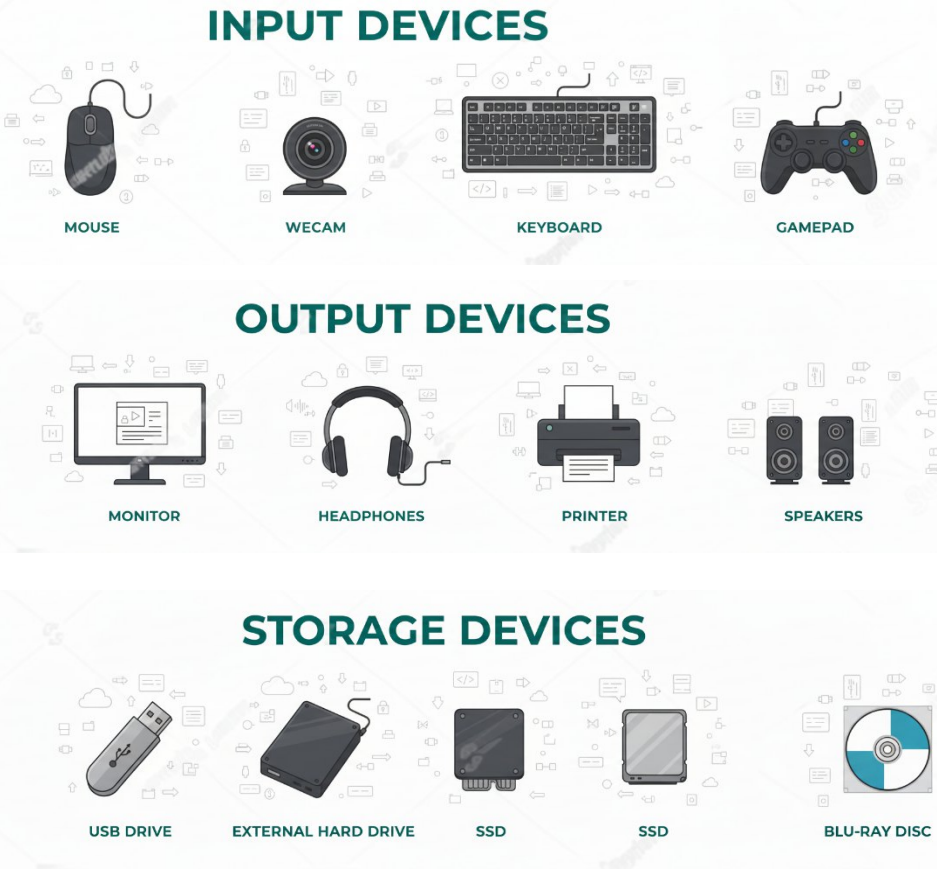
Core Features of Computers

- ✓ Speed: Processes data and performs calculations far faster than humans.
- ✓ Accuracy: Executes tasks with high precision when properly programmed.
- ✓ Storage: Holds vast amounts of data, from documents to multimedia.
- ✓ Connectivity:Links to networks and the internet for communication and sharing.
- ✓ Versatility:Performs a wide range of tasks across different industries.
- ✓ Automation:Automates repetitive tasks, improving efficiency.

Key Components of Computer



physical parts of a



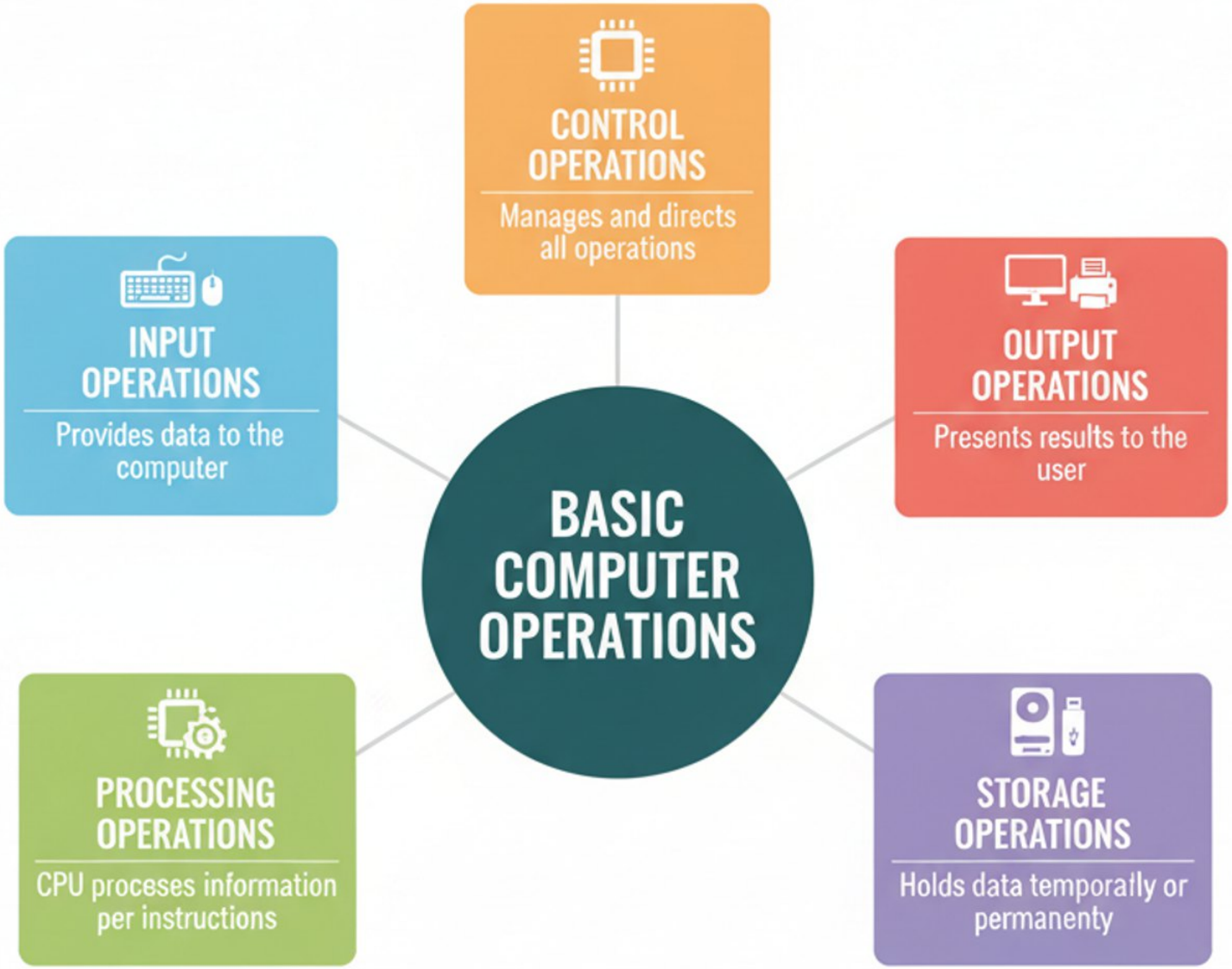


instructions that direct the hardware to perform specific tasks

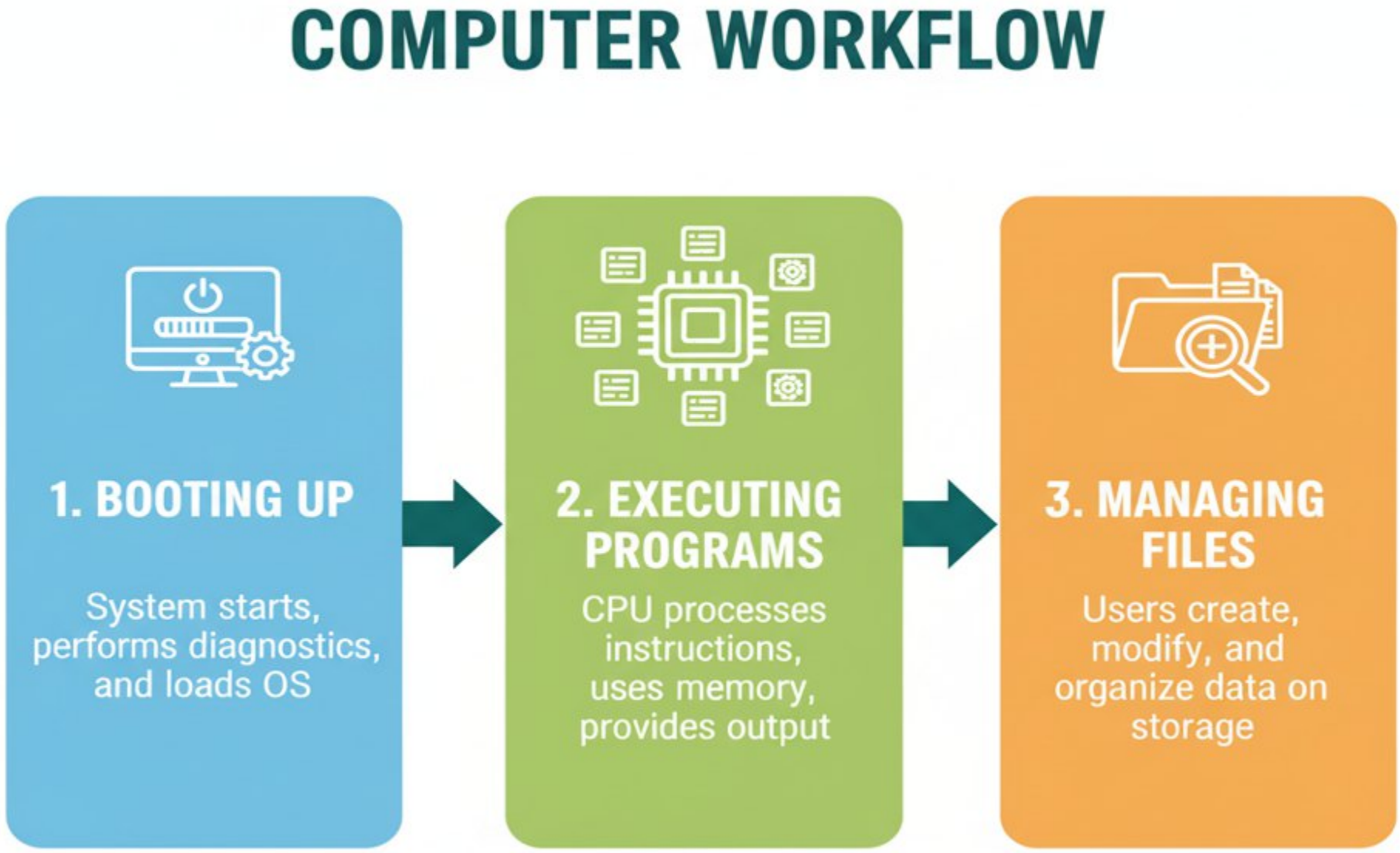




Basic Computer Operations



Computer Workflow

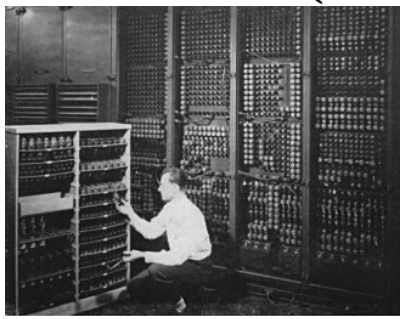


Events after you press the power button

- ✓ Power-On Self Test (POST): The computer performs a self-check to ensure all hardware components are functioning correctly.
- ✓ Boot Loader: The computer loads a small program called the bootloader, which in turn loads the operating system into memory.
- ✓ Operating System: The operating system initializes various hardware components and prepares the system for use.
- ✓ User Interface: Finally, the user interface (such as Windows Desktop or MacOS Finder) is displayed, allowing you to interact with the computer.

Generations of Computers

First Generation (1940s–1950s)



- ✓ Used vacuum tubesand magnetic drums
- ✓ Large, expensive, and heat-intensive.
- ✓ Examples:ENIAC, UNIVAC I

Second Generation (1950s–1960s)



- ✓ Replaced vacuum tubes with transistors → smaller & reliable.
- ✓ Introduction of assembly language & high-level languages (Fortran, COBOL).

Second Generation (1960s–1970s)



- ✓ Integrated Circuits (ICs) replaced transistors.
- ✓ Smaller, faster, more reliable.
- ✓ Features: Operating systems, multiprocessing

Fourth Generation (1970s–1980s)



- ✓ Microprocessors → PCs became common.
- ✓ Innovations: GUIs, software applications, internet access

Fifth Generation (1980s–Present)



- ✓ Focus on AI, machine learning, quantum computing.
- ✓ Features: Parallel processing, intuitive interfaces, mobile & connected devices.

Types of Computers (by size)



Supercomputers: Extremely powerful, perform trillions of calculations/sec for tasks like simulations and data analysis. Examples: IBM Summit, Sunway TaihuLight



Mainframe Computers: Large, powerful systems for handling bulk data processing, transactions, and critical applications. Examples: IBM zSeries, Unisys ClearPath.



Minicomputers: Mid-sized computers for small businesses and departmental tasks. Examples: DEC PDP-11, IBM AS/400.



Microcomputers: Personal computers for individual use, like desktops and laptops. Examples: Dell Inspiron, Apple MacBook.



Workstation Computers: High-performance PCs for technical or scientific tasks. Examples: HP Z Workstation, Dell Precision.



Embedded Computers: Specialized computers built into devices to perform specific tasks. Examples: Arduino, Raspberry Pi.

Types of Computers (by Data Type)

BY DATA TYPE



ANALOG COMPUTER

Processes continuous, varying data (.temperature, pressure, voltage).



DIGITAL COMPUTER

Processes discrete data represanted in binary (0s and Ins)



HYBRID COMPUTER

Combines speed of of anllog with accuracy of digital.

Types of Computers (by Purpose)

- BY PURPOSE

GENERAL PURPOSE COMPUTER



Designed to perform a wide variety tasks by running different software applications (eg, PC, laptop).

SPECIAL PURPOSE COMPUTER



Built for one specific, dedicated function or a very limited set tasks (e.g, car's computer, mickowave).

Limitations of Computer



No IQ

Can not think creatively or understand without input.



Vulnerability to Cyber Threats

Exposed to cyber threats like hacking, viruses, and malware, risking data integrity, operations, and finances.



Lack of Decision Making

They can analyze data but cannot make subjective or moral judgments.



Ethical and Moral Reasoning

computers cannot reason ethically or morally; they can only follow human-defined guidelines.



Human Dependency

Need Human programming, instructions and data input.



No feelings

Can not understand or express human feelings/empathy.